

Program 4

Design and implement C/C++ Program to find shortest paths from a given vertex in a weighted connected graph to other vertices using Dijkstra's algorithm.

```
#include<stdio.h>
```

```
void main()
```

```
{
    int i,j,a[10][10],s[10],n,d[10],v,k,min,u;

    printf("enter the no of vertices:=>");
    scanf("%d",&n);

    printf("enter the cost matrix\n");
    printf("enter 999 for no edges between vertices\n");
    for(i=1;i<=n;i++)
        for(j=1;j<=n;j++)
            scanf("%d",&a[i][j]);

    printf("enter the source vertex\n");
    scanf("%d",&v);

    for(i=1;i<=n;i++)
    {
        s[i]=0;
        d[i]=a[v][i];
    }

    d[v]=0;
    s[v]=1;
    for(k=2;k<=n;k++)
    {
        min=999;
        for(i=1;i<=n;i++)
            if(s[i]==0&&d[i]<min)
            {
                min=d[i];
                u=i;
            }
        s[u]=1;
        for(i=1;i<=n;i++)
            if(s[i]==0)
            {
                if(d[i]>d[u]+a[u][i])
                    d[i]=d[u]+a[u][i];
            }
    }

    printf("the shortest distance from %d is\n",v);
    for(i=1;i<=n;i++)
        printf("\n%d->%d=%d",v,i,d[i]);
}
```

Sample Input and Output:

enter the no of vertices:=>5

enter the cost matrix

enter 999 for no edges between vertices

999 3 999 7 999

3 999 4 2 999

999 4 999 5 6

7 2 5 999 4

999 999 6 4 999

enter the source vertex

1

the shortest distance from 1 is

1->1=0

1->2=3

1->3=7

1->4=5

1->5=9